

The 1000 m² Mars Rubber-Glass House — Version 2.0

A Comprehensive Design for Long-Duration Habitation, Closed-Loop Life Support, and Operational Crew Routines

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Abstract

This paper presents the Version 2.0 design of the 1000 m² Mars Rubber-Glass House, a comprehensive upgrade from the structural feasibility study (v1, June 2024) into a fully specified habitat for long-duration human habitation. Building on the Design Improvements v2 proposals (increased height, thermal zoning, radiation shielding, expanded agriculture, energy storage, and crew privacy), v2.0 integrates five major new systems: (1) a complete closed-loop waste-to-resource cycle converting human excreta, food waste, and atmospheric by-products into fertiliser, water, and chemical feedstocks; (2) structured daily, weekly, monthly, and annual crew routines modelled on operational experience from Biosphere 2, MARS-500, HI-SEAS, ISS, and Antarctic stations; (3) an in-situ resource utilisation (ISRU) strategy for manufacturing construction materials, propellants, and speciality chemicals from Martian regolith and atmosphere; (4) a medical and psychological support framework informed by confined-environment research; and (5) a simulation and validation methodology following the cycle Investigation → Model → Simulation → Validation → Scientific Assessment. The result is a design that transforms the Rubber-Glass House from a shelter into a self-sustaining micro-civilisation capable of supporting four crew members indefinitely.

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1. Introduction

1.1 Background

The original Mars Rubber-Glass House (v1) established the structural and material foundation: a 1000 m² (50 m × 20 m) single-storey habitat with silicone rubber foundations, a titanium/Inconel frame, borosilicate glass walls, rooftop solar panels, and a single airlock — all at a materials cost of approximately \$1.02 M (DOI: 10.5281/zenodo.16493055).

The Design Improvements v2 document identified critical gaps in the v1 design and proposed eight upgrades:

1. Interior height increase from 3 m to 5 m
2. Three thermal zones (warm, temperate, cool)
3. Closed-loop water management
4. Multi-layered radiation shielding
5. Agricultural system expansion with vertical farming
6. Airlock redundancy (2 + emergency hatch)
7. Energy storage and redundancy (100 kWh battery + RHU)
8. Acoustics and crew privacy (individual cabins)

1.2 Scope of Version 2.0

Version 2.0 accepts all eight v2 proposals as baseline and extends the design into a **complete operational habitat** by addressing five previously unspecified domains:

- **Closed-loop waste-to-resource cycling** — from human excreta and food waste to fertiliser, water, and chemical feedstocks
- **Structured crew routines** — daily, weekly, monthly, and annual schedules informed by Biosphere 2, MARS-500, HI-SEAS, ISS, and Antarctic station operations
- **In-situ resource utilisation (ISRU)** — manufacturing construction materials, propellants, and speciality chemicals from Martian resources
- **Medical and psychological support** — informed by confined-environment psychology research
- **Simulation and validation methodology** — a rigorous cycle of Investigation → Model → Simulation → Validation → Scientific Assessment

1.3 Lessons from Earth Isolation Analogues

The v2.0 design is informed by operational experience from five key analogues:

Analogue	Duration	Crew	Key Lessons
Biosphere 2 (1991-1993)	2 years	8	Agriculture produced 83% of diet; O ₂ dropped from 20.9% to 14.5%; CO ₂ management critical; crew lost 16% body weight; crew factions formed; constructed wetlands recycled sewage
MARS-500 (2010-2011)	520 days	6	Sleep disruption in 4/6 crew; activity levels collapsed; circadian rhythm disruption; no interpersonal conflicts reported officially

Analogue	Duration	Crew	Key Lessons
HI-SEAS IV (2015-2016)	366 days	6	Food morale critical; water system failures occurred; communication delay simulation; crew dynamics under study
Concordia Station (ongoing)	9 months/winter	13	Extreme isolation; hypobaric hypoxia at 3,233 m; circadian disruption; analogue for deep-space missions
ISS (ongoing)	6-12 months	3-6	Noise and privacy are top stressors; structured schedules essential; exercise prevents bone/muscle loss; WRS achieves 93% water recovery

These lessons directly inform every section of this document.

2. Structural Design — Consolidated v2 Baseline

2.1 Envelope Dimensions

Parameter	v1	v2.0
Footprint	50 m × 20 m = 1,000 m ²	50 m × 20 m = 1,000 m ²
Interior height	3.0 m	5.0 m
Interior volume	3,000 m ³	5,000 m³
Foundation wall	140 m perimeter × 0.5 m × 0.1 m	unchanged
Mezzanine level	—	~400 m² at 2.5 m height
Total usable floor area	1,000 m ²	~1,400 m² (ground + mezzanine)

2.2 Structural Materials

Component	Material	v2.0 Mass
Floor slab	Silicone rubber (10 cm)	107 m ³
Foundation wall	Silicone rubber	~7 m ³
Partition foam	Silicone foam insulation	~5 m ³
Frame	Titanium / Inconel	~8,500 kg
Glazing (walls)	Borosilicate glass + insulation coating	~700 m ² (~16,700 kg)

Component	Material	v2.0 Mass
Roof	Solar panels (1,000 m ²) + regolith layer (50 cm)	~10,000 kg panels + ~800,000 kg regolith
Airlocks	2 × Titanium/Inconel + borosilicate	~2,000 kg
Emergency hatch	1 × 1 m × 1 m reinforced	~200 kg

2.3 Thermal Zoning

Three climate zones separated by lightweight silicone foam partition walls (STC ≥ 35):

SOLAR ROOF (1000 m ²) + 50 cm Martian regolith layer		
WARM ZONE ~350 m ² 22–24°C Living quarters, kitchen, cabins, quiet rm [Airlock NORTH]	TEMPERATE ZONE ~250 m ² 18–20°C Labs, workshops, machinery room, medical bay, ISRU processing, waste processing	COOL ZONE ~400 m ² 12–16°C Agriculture, vertical farming, algae bioreactors, aquaponics, composting [Airlock SOUTH]
MEZZANINE LEVEL (above warm + temperate zones) ~400 m ² : 4 crew cabins (48 m ²), quiet room (12 m ²), storage, crew recreation, vertical farm upper tiers		

Energy savings: Estimated 20–30% reduction in heating load compared to uniform heating, as the agricultural zone (40% of floor area) operates at 12–16 °C with day/night cycling.

3. Closed-Loop Life Support System (CLLSS)

The v2.0 habitat operates as a near-closed bioregenerative life-support system. The goal is **≥ 95% closure** for water and **≥ 80% closure** for food, following the Biosphere 2 model but improved with 30 years of subsequent research.

3.1 Atmospheric Management

Parameter	Target	Method
O ₂ partial pressure	21.0 kPa (± 0.5)	Photosynthesis + electrolysis backup
CO ₂ partial pressure	0.04-0.10 kPa	Plant uptake + scrubber fallback
Total pressure	70 kPa	Pressure vessel + relief valves
Relative humidity	40-60%	Condensation + dehumidification
Temperature (warm zone)	22-24 °C	Active heating + thermal mass
Trace gas removal	< SMAC limits	Activated carbon + catalytic oxidiser

Lesson from Biosphere 2: O₂ declined from 20.9% to 14.5% over 16 months due to microbial respiration in organic-rich soils and CO₂ absorption by unsealed concrete. V2.0 mitigates this through: - Sealed silicone rubber floor preventing soil-concrete reactions - Dedicated algae bioreactors (40 m²) optimised for O₂ production - Backup electrolysis unit (splitting water → H₂ + O₂) - Real-time atmospheric monitoring with automated adjustment

3.2 Water Management

Water budget per crew member (with recycling):

Source	Volume/day	Notes
Drinking water	2.5 L	Potable grade
Food preparation	1.0 L	
Personal hygiene	6.0 L	Shower (Navy-style, 2 min)
Laundry	5.0 L	Weekly, shared
Agriculture irrigation	8.0 L	Recirculating hydroponic
Toilet flush / processing	2.5 L	Vacuum-assist
Total per person	25 L/day	
Total for 4 crew	100 L/day	

Water recycling system (modelled on ISS WRS):

1. **Humidity condensate recovery** — Air handlers collect ~4 L/day per person from transpiration and respiration
2. **Greywater filtration** — Shower, laundry, and kitchen water filtered through biofilm reactor + activated carbon + UV sterilisation
3. **Urine processing** — Vapour compression distillation (VCD) achieving 93% recovery rate
4. **Plant transpiration recovery** — The agricultural zone alone recycles ~60 L/day through plant transpiration → condensation
5. **Underfloor reservoir** — 10 m³ emergency reserve integrated into silicone rubber floor slab (inherently watertight)

Target water recovery rate: ≥ 93% (ISS demonstrated), aspiration ≥ 97% with plant transpiration loop.

3.3 Waste-to-Resource Cycling

This is a **key innovation of v2.0**. Every waste stream is treated as a resource input:

3.3.1 Human Excreta Processing

Waste Stream	Daily Volume (4 crew)	Processing	Output
Urine	~6 L	VCD → water; urea extraction → nitrogen fertiliser	~5.5 L water + ~30 g urea/day
Faeces	~0.6 kg	Thermophilic composting (60–70 °C, 14 days) → vermicomposting (21 days)	~0.15 kg stabilised compost/day
Greywater sludge	~2 L	Biofilm digestion → nutrient extraction	Phosphorus + potassium salts

Composting process: 1. **Collection** — Vacuum toilet separates urine and faeces at source 2. **Pre-treatment** — Faeces mixed with crop residue (C:N ratio 25:1–30:1) 3. **Thermophilic composting** — Enclosed reactor at 60–70 °C for 14 days; kills pathogens (Salmonella, E. coli, helminth ova) 4. **Vermicomposting** — *Eisenia fetida* (red wiggler worms) process compost for additional 21 days, producing nutrient-rich vermicast 5. **Maturation** — 30 days aerobic curing 6. **Application** — Distributed to agricultural beds as slow-release fertiliser

Safety: The thermophilic stage exceeds WHO guidelines for pathogen destruction (≥ 55 °C for ≥ 14 days). Compost is tested for faecal coliforms before agricultural use.

3.3.2 Food Waste Processing

Waste Type	Daily Volume (4 crew)	Processing	Output
Crop residue (stems, roots, leaves)	~3 kg	Shredding → composting or biochar	Compost or soil amendment
Kitchen waste (peels, scraps)	~0.5 kg	Vermicomposting or anaerobic digestion	Vermicast or biogas (CH_4)
Inedible biomass	~1 kg	Pyrolysis (400 °C, O_2 -free)	Biochar (soil carbon) + syngas

3.3.3 Atmospheric Waste Cycling

Gas	Source	Processing	Output
CO_2 (exhaled)	~1 kg/person/day	Plant photosynthesis + Sabatier reactor	O_2 + biomass; or CH_4 + H_2O
CH_4 (from digestion)	Trace	Catalytic oxidation	CO_2 + H_2O (fed back to plants)
NH_3 (from composting)	Trace	Biofilter (nitrifying bacteria)	NO_3^- (liquid fertiliser)

Gas	Source	Processing	Output
VOCs (from materials)	Trace	Activated carbon adsorption	Cleaned air

3.3.4 Chemical and Raw Material Recovery The closed-loop system also produces speciality outputs:

Product	Source	Process	Use
Nitrogen fertiliser (urea)	Urine	Concentration + crystallisation	Agriculture
Phosphorus salts	Greywater + compost leachate	Struvite precipitation (MgNH_4PO_4)	Agriculture
Potassium salts	Plant ash + compost leachate	Leaching + evaporation	Agriculture
Biochar	Crop residue pyrolysis	Pyrolysis at 400 °C	Soil amendment, water filtration
Methane (CH_4)	Anaerobic digestion of food waste	Mesophilic digestion (35 °C)	Fuel for heating or Sabatier input
Oxygen (O_2)	Water electrolysis backup	PEM electrolysis	Atmosphere supplement
Hydrogen (H_2)	Water electrolysis	PEM electrolysis	Sabatier reactor input

3.4 Nutrient Mass Balance

A simplified annual nutrient balance for 4 crew:

Nutrient	Input Required (kg/yr)	From Waste Recycling (kg/yr)	From ISRU (kg/yr)	Deficit/Surplus
Nitrogen (N)	40	30 (from urine + composting)	10 (from atmospheric N_2 fixation)	Balanced
Phosphorus (P)	8	6 (from struvite + compost)	2 (from regolith extraction)	Balanced
Potassium (K)	15	10 (from plant ash + compost)	5 (from regolith)	Balanced
Carbon (C)	200	180 (from CO_2 + biomass cycling)	20 (from atmospheric CO_2)	Balanced
Water (H_2O)	36,500 L	34,675 L (95% recycling)	1,825 L (from ice extraction)	Balanced

4. Agricultural System

4.1 Design Parameters

Parameter	v1	v2.0
Base agricultural area	200 m ² (50 m ² /person)	400 m² (100 m²/person)
Vertical tiers	1	3 tiers in cool zone
Effective growing area	200 m ²	~1,200 m²
Algae bioreactors	20 m ²	40 m²
Aquaponics system	—	30 m² fish tanks + grow beds
LED supplemental lighting	—	Full-spectrum, 18 h photoperiod
Growing method	Soil-based	Hybrid: hydroponics + soil beds

4.2 Crop Selection and Yield Targets

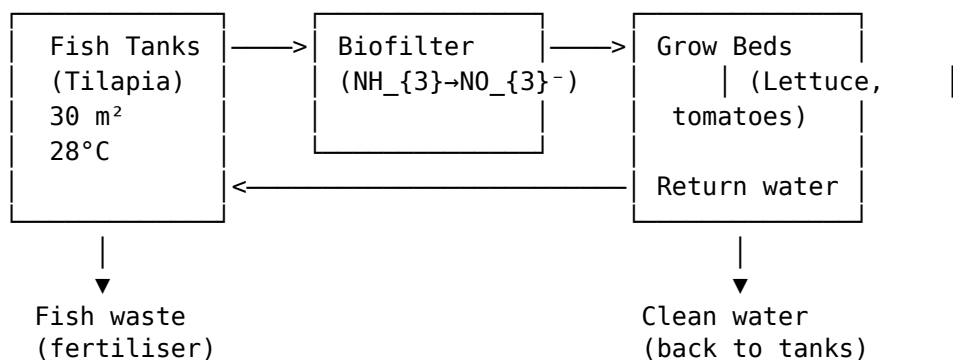
Crop selection is optimised for caloric density, growth speed, nutritional completeness, and CO₂-to-O₂ conversion efficiency:

Crop	Area (m ²)	Yield (kg/m ² /yr)	Calories (kcal/kg)	Annual kcal	Role
Potatoes	200	15	770	2,310,000	Staple carbohydrate
Soybeans	150	3	1,460	657,000	Protein + oil
Wheat	200	4	3,390	2,712,000	Staple grain
Rice	100	5	1,300	650,000	Staple grain
Let-tuce/greens	150	20	150	450,000	Vitamins + minerals
Tomatoes	100	30	180	540,000	Vitamins + morale
Sweet potatoes	100	12	860	1,032,000	Carbohydrate + vitamin A
Spirulina (algae)	40	50 (dry)	2,900	5,800,000	Protein + O ₂ production
Tilapia (aquaponics)	30	8 (fish/m ² /yr × 0.5 kg)	960	115,200	Animal protein
Total	1,070			~14,266,200	
Required (4 crew × 2,500 kcal/day × 668 sols)				~6,680,000	

The system produces roughly **2× the minimum caloric requirement**, providing a safety buffer consistent with Biosphere 2’s lesson that agricultural shortfalls are the most dangerous failure mode (crew lost 16% body weight despite “the highest producing farm in the world”).

4.3 Aquaponics Integration

A closed-loop fish-plant system provides protein diversification and nutrient cycling:



- **Fish species:** Tilapia (*Oreochromis niloticus*) — hardy, fast-growing, omnivorous, previously used in Biosphere 2's rice-azolla pond system
- **Feed:** Spirulina + crop residue + insect meal (black soldier fly larvae fed on food waste)
- **Yield:** ~120 kg fish/year (30 g protein/day for 4 crew)

4.4 Lighting System

Mars receives only ~43% of Earth's solar irradiance (590 W/m² peak at Mars vs. 1,361 W/m²). During global dust storms (lasting 2-8 weeks), solar input can drop to 10-20% of nominal.

Supplemental LED array: - Full-spectrum LEDs (red 660 nm + blue 450 nm + white) — optimised for photosynthetically active radiation (PAR) - Power requirement: ~50 W/m² × 1,200 m² effective area × 18 h/day = **1,080 kWh/day** at full power - During clear conditions: LEDs supplement natural sunlight (reduce to ~30% power) - During dust storms: LEDs operate at 100% to maintain crop growth - Light scheduling: 18 h light / 6 h dark cycle (optimised for most crop species)

5. Energy System

5.1 Power Generation

Source	Capacity	Annual Output	Notes
Rooftop solar panels	1,000 m² × ~150 W/m² peak	~973,392 kWh/Mars year	Dust storm degradation: 10-20%
Radioisotope heater units (RHU)	3 × 1 kW thermal	~26,280 kWh thermal/year	Backup heat only

5.2 Power Budget

System	Daily Consumption (kWh)	Priority
Life support (atmosphere, temperature)	120	Critical
Agricultural lighting (LEDs)	400-1,080	High
Water recycling	30	Critical
Waste processing	15	Medium
ISRU processing	50	Low
Crew systems (lighting, cooking, comms)	40	Medium
Laboratory equipment	20	Low
Total nominal	~675-1,355	

System	Daily Consumption (kWh)	Priority
Available (clear day)	~1,460	973,392 ÷ 668 sols

5.3 Energy Storage and Load Management

- **Battery bank:** 200 kWh LFP (upgraded from v2 proposal of 100 kWh)
 - Provides ~72 hours of critical systems (life support + water + minimal agriculture lighting)
 - Weight: ~1,600 kg
- **Load prioritisation during dust storms:**
 1. Life support (atmosphere, temperature) — always on
 2. Water recycling — always on
 3. Agriculture lighting — reduced to 30% (critical crops only)
 4. Crew systems — reduced
 5. Laboratory + ISRU — suspended

6. Radiation Shielding

6.1 Radiation Environment

Mars surface radiation (MSL/RAD data): - GCR dose rate: ~0.67 mSv/day (~244 mSv/year) - SPE events: up to 50 mSv in hours (rare but dangerous) - NASA career limit: 600 mSv (age/sex dependent)

6.2 Multi-Layer Shielding Design

Layer	Location	Material	Thickness	Dose Reduction
Regolith blanket	Roof (above solar panels)	Martian regolith	50 cm (~80 g/cm ²)	~50% GCR
Water walls	Side walls (double-glass)	Water	10-20 cm	~20-30% GCR + excellent neutron shielding
Hydrogen-rich polyethylene	Storm shelter walls	HDPE panels	10 cm	Additional ~15% for shelter
Regolith berm	Storm shelter	Compacted regolith	≥ 2 m overhead	~90% total for shelter

6.3 Storm Shelter

- **Location:** Central portion of temperate zone
- **Area:** ~20 m² (5 m × 4 m)
- **Capacity:** 4 crew for up to 72 hours
- **Shielding:** ≥ 2 m regolith overhead + water walls on all sides
- **Equipment:** Emergency life support, 200 L water, food rations (72 h), medical kit, communications terminal
- **Effective dose rate in shelter:** < 0.05 mSv/day

6.4 Combined Shielding Performance

Location	Dose Rate (mSv/day)	Annual Dose (mSv)
Unshielded Mars surface	0.67	244
Inside habitat (general)	~0.20-0.25	~73-91
Inside storm shelter	< 0.05	< 18
NASA career limit reference	—	600

A 2-year stay at ~0.22 mSv/day = ~160 mSv — well within NASA career limits.

7. Crew Routines — Operational Schedules

7.1 Design Philosophy

Structured schedules are essential for crew health, morale, and productivity. Research from all major analogues confirms:

- **ISS:** Highly structured 16-hour work / 8-hour sleep schedule prevents drift
- **MARS-500:** Unstructured time led to crew becoming progressively inactive; 4 of 6 developed sleep disorders
- **Biosphere 2:** Crew faction formation was exacerbated by disagreements over daily priorities
- **HI-SEAS:** Crew performed 40 tasks/week; structured activities maintained engagement
- **Concordia:** Circadian disruption during polar night requires strict light/dark scheduling

7.2 Daily Schedule (Sol = 24 h 37 min)

The daily schedule follows a **16/8 rhythm** (16 h active, 8 h sleep), with solar time adjusted for Mars's slightly longer day:

Time (Mars Local)	Duration	Activity	Zone
06:00-06:30	30 min	Wake-up, personal hygiene	Warm
06:30-07:00	30 min	Morning medical check (weight, vitals, mood log)	Warm
07:00-07:45	45 min	Breakfast (communal)	Warm
07:45-08:00	15 min	Daily briefing + task assignment	Temperate
08:00-10:00	2 h	Agricultural maintenance — harvest, planting, pruning, nutrient checks	Cool
10:00-10:15	15 min	Break	—

Time (Mars Local)	Duration	Activity	Zone
10:15-12:15	2 h	Science / Laboratory work — experiments, data analysis, Earth comms	Temperate
12:15-13:00	45 min	Lunch (communal)	Warm
13:00-13:30	30 min	Siesta / personal time	Warm
13:30-15:30	2 h	Maintenance + Engineering — systems checks, repairs, ISRU	Temperate
15:30-15:45	15 min	Break	—
15:45-17:00	1.25 h	EVA or training (2 crew, alternating pairs) or indoor exercise	Various
17:00-18:00	1 h	Mandatory exercise — resistance training + cardio	Warm
18:00-18:30	30 min	Personal hygiene + shower	Warm
18:30-19:15	45 min	Dinner (communal)	Warm
19:15-20:00	45 min	Evening briefing — log entries, next-sol planning, Earth video uplink	Temperate
20:00-22:00	2 h	Personal time — recreation, reading, video calls, hobbies	Warm / Cabins
22:00-06:00	8 h	Sleep (individual cabins, blue-light-free)	Cabins

Critical design features: - **Three communal meals** — Biosphere 2 and HI-SEAS both found shared meals are the most important social bonding activity - **Mandatory exercise** — ISS protocol: 2 h/day (1 h resistance + 1 h cardio) prevents bone loss (~1.5%/month) and muscle atrophy - **Morning medical check** — Body weight, blood pressure, pulse oximetry, mood self-assessment on a 1-10 scale (MARS-500 found early detection of psychological issues is critical) - **Personal time** — Minimum 2 h/day unscheduled time (ISS finding: overscheduling causes crew rebellion — see Skylab 4 “strike”) - **Blue-light curfew** — No blue-spectrum lighting after 20:00 (Concordia research shows circadian disruption is a major issue)

7.3 Weekly Schedule

Each sol has a primary focus. The week is structured around a **6-day work / 1-day rest** cycle (consistent with ISS operations):

Sol	Focus	Special Activities
Sol 1 (Monday)	Agriculture	Deep harvest; crop rotation planning; seed germination checks; compost turning
Sol 2 (Tuesday)	Science	Laboratory experiments; geological sample analysis; Earth data downlink
Sol 3 (Wednesday)	Engineering	Systems maintenance; filter changes; pressure checks; ISRU operations
Sol 4 (Thursday)	EVA Day	Pair EVA (2 crew, 4 h); regolith collection; exterior inspection; sensor deployment
Sol 5 (Friday)	Medical + Training	Full medical examinations; emergency drills; skills cross-training
Sol 6 (Saturday)	Housekeeping	Deep cleaning; laundry; waste processing check; inventory review
Sol 7 (Sunday)	Rest Day	No scheduled work; personal projects; recreation; crew social activities; Earth family calls

Weekly mandatory activities: - 1 × EVA sortie (2 crew, minimum 4 h) - 1 × emergency drill (fire, depressurisation, medical, communication loss — rotating) - 1 × crew meeting (open discussion of issues, morale check) - 1 × deep-clean of all communal areas - 1 × water quality analysis (microbial + chemical) - 1 × atmospheric composition verification (manual cross-check of sensors)

7.4 Monthly Schedule

Monthly activities layer onto the weekly cycle:

Week	Monthly Activity
Week 1	Crop Cycle Review — Assess yields vs. targets; adjust planting schedules; replace depleted nutrient solutions; recalibrate pH sensors
Week 2	Systems Deep Maintenance — Full inspection of all life support subsystems; replace HEPA filters; clean water filtration membranes; test backup power systems
Week 3	Medical Review — Comprehensive health assessment: blood work, bone density scan (portable DXA), psychological evaluation (standardised questionnaire + video call with Earth psychologist); vision test (ICP monitoring)

Week	Monthly Activity
Week 4	Science Report + Planning — Monthly science report compiled and transmitted to Earth; review of experimental results; plan next month's experiments; inventory reconciliation

Monthly metrics tracked:

Category	Metrics
Agriculture	Crop yield (kg); O ₂ production rate; CO ₂ fixation rate; water consumption; pest/disease incidence
Life Support	Water recovery rate (%); air composition (O ₂ , CO ₂ , trace gases); power consumption; waste processing throughput
Medical	Body mass; bone density; blood pressure; HbA1c; immune markers; sleep quality (actigraphy); mood trends
Engineering	System uptime (%); repair frequency; consumable inventory; ISRU output
Psychological	Crew cohesion (sociometry); stress indicators; conflict incidents; morale index

7.5 Annual Schedule (Mars Year = 668.6 Sols)

The annual schedule aligns with **Martian seasons** (Mars has a 25.19° axial tilt, creating seasons similar to Earth but ~twice as long):

Period (Sols)	Season	Focus	Key Events
Sol 1-167	Northern Spring	Foundation Phase	Major crop planting; exterior construction/repair (calm weather); scientific survey campaigns; EVA frequency: 2/week
Sol 168-334	Northern Summer	Productivity Phase	Peak harvest; maximum solar energy; ISRU production; long-range EVA expeditions; lowest dust storm risk
Sol 335-501	Northern Autumn	Preparation Phase	Harvest storage; system winterisation; dust storm preparation; battery charge maximisation; reduced EVA

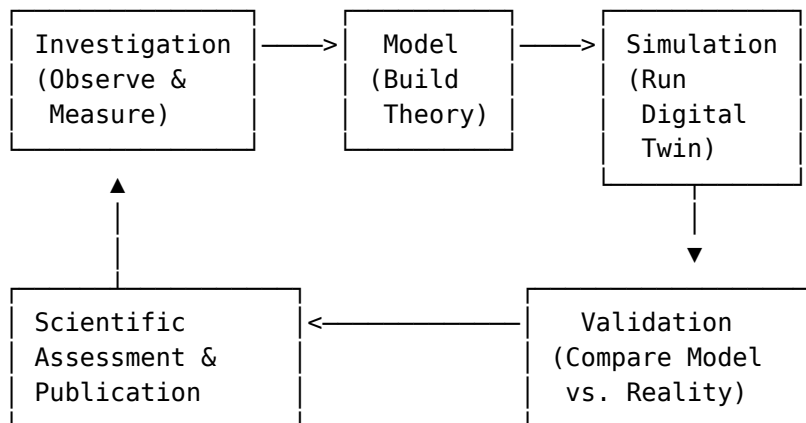
Period (Sols)	Season	Focus	Key Events
Sol 502-668	Northern Winter	Conservation Phase	Reduced solar input; dust storm season; focus on indoor science, maintenance, training; LED agriculture at full power; storm shelter readiness

Annual milestones:

Sol	Milestone
Sol 1	New Year ceremony; annual goals setting; crew role rotation review
Sol 50	First major harvest review
Sol 167	Spring review; publication of first seasonal science report
Sol 334	Mid-year comprehensive medical examination; psychosocial assessment
Sol 400	Dust storm preparedness drill (full 72-h shelter exercise)
Sol 500	Winter solstice celebration; crew appreciation event
Sol 600	Pre-annual-review system audit
Sol 668	Annual review; data compilation; next-year planning

7.6 Simulation and Validation Methodology

Each operational system follows a rigorous five-stage cycle:



Applied to each domain:

Domain	Investigation	Model	Simulation	Validation	Assessment
Agriculture	Measure crop yields, growth rates, nutrient uptake weekly	Build growth models: $\text{yield} = f(\text{light, temperature, nutrients, CO}_2)$	Run digital twin with varied inputs (dust storm scenarios, nutrient deficiency)	Compare predicted vs. actual yields monthly	Publish seasonal agricultural performance reports
Water cycle	Monitor daily: flow rates, quality parameters, recovery rate	Mass balance model: inputs, losses, recycling efficiency	Simulate failure modes (pump failure, membrane fouling)	Compare model predictions vs. sensor data	Annual water cycle performance paper
Waste cycling	Track compost maturation, nutrient content, microbial activity	Nutrient flow model: N, P, K balance across all waste streams	Simulate optimal C:N ratios, temperature profiles, retention times	Lab-test compost samples against model predictions	Publish waste-to-resource conversion efficiency data
Energy	Log power generation, consumption, storage levels daily	Energy balance model incorporating seasonal solar variation	Simulate dust storm scenarios (2, 4, 8-week duration)	Compare actual power reserves vs. model during real storms	Annual energy system performance report
Crew health	Daily vitals, weekly mood surveys, monthly bloods	Crew health model: $\text{fitness} = f(\text{exercise, nutrition, sleep, stress})$	Predict health trajectories under reduced exercise / poor sleep	Compare with actual medical data	Contribute to interplanetary medicine literature
Structural	Monitor pressure, temperature gradients, material fatigue sensors	FEA model of structure under Martian conditions	Simulate extreme events (meteorite impact, dust storm loading)	Compare sensor data vs. model predictions	Engineering performance reports

8. In-Situ Resource Utilisation (ISRU)

8.1 Available Martian Resources

Resource	Source	Abundance	Processing
SiO₂ (silicon dioxide)	Regolith (~45% SiO ₂)	Abundant	Glass, silicone rubber, aerogel
Fe₂O₃ (iron oxide)	Regolith (~18% Fe ₂ O ₃)	Abundant	Iron/steel for tools, structures
CO₂	Atmosphere (95.3%)	Effectively unlimited	Sabatier → CH ₄ + O ₂ ; feedstock for plant growth
N₂	Atmosphere (2.7%)	Limited but extractable	Fertiliser (Haber-Bosch requires H ₂ + N ₂)
H₂O (ice)	Subsurface ice deposits	Variable by location	Water, H ₂ (electrolysis), O ₂
Ar (argon)	Atmosphere (1.6%)	Available	Inert gas for welding, fire suppression
Al₂O₃, TiO₂, MgO	Regolith	Minor components	Aluminium, titanium extraction (long-term)

8.2 ISRU Processing Chain

Phase 1 (from Sol 1): Atmospheric Processing - CO₂ collection + Sabatier reaction: CO₂ + 4H₂ → CH₄ + 2H₂O - Produces methane (fuel/heating) + water - Water electrolysis: 2H₂O → 2H₂ + O₂ - Oxygen for atmosphere; hydrogen recycled to Sabatier

Phase 2 (from Sol 100): Regolith Processing - Regolith collection by robotic systems (DOI: 10.5281/zenodo.15455156) - Silica extraction for glass repair and silicone rubber production - Iron oxide reduction for basic metal tools - Regolith sintering for construction bricks

Phase 3 (from Sol 300): Chemical Manufacturing - Haber-Bosch process (N₂ + H₂ → NH₃) — requires significant energy but enables fertiliser independence - Perchlorate reduction from regolith — detoxification of Martian soil for agriculture - Water extraction from subsurface ice deposits

8.3 ISRU Facility Requirements

Equipment	Mass (kg)	Power (kW)	Location
CO ₂ collector + compressor	200	2	Exterior
Sabatier reactor	150	1	Temperate zone
Water electrolysis unit	100	5	Temperate zone
Regolith processing oven	300	8	Exterior (shielded)
Haber-Bosch mini-reactor	200	3	Temperate zone
Total	950	19 peak	

9. Medical and Psychological Support

9.1 Medical Facility

Located in the temperate zone, the medical bay occupies ~15 m²:

Equipment	Purpose
Portable ultrasound	Diagnostic imaging (replaces X-ray — no radiation source needed)
Blood analyser (point-of-care)	CBC, metabolic panel, HbA1c, CRP, electrolytes
Portable DXA scanner	Bone density monitoring (critical for Mars gravity: 0.38 g)
Surgical kit (minor)	Wound repair, tooth extraction, abscess drainage
Pharmacy cabinet	Antibiotics, analgesics, anaesthetics, cardiac drugs, antidepressants
Defibrillator + AED	Cardiac emergency
Telemedicine terminal	Video consultation with Earth (4–24 min delay)
Psychological assessment tools	PHQ-9 (depression), GAD-7 (anxiety), PSQI (sleep quality), crew sociometry

9.2 Psychological Support Framework

Drawing on lessons from all analogues:

Stressor	Mitigation	Evidence
Isolation	Weekly Earth video calls; crew social events; varied daily activities	MARS-500: crew spent increasing time alone; ISS: scheduled family calls essential
Confinement	5 m ceiling height; glass walls with exterior views; varied thermal zones; EVA opportunities	Submarines/Antarctic: perceived spaciousness correlates with morale
Interpersonal conflict	Conflict resolution training pre-mission; private cabins; quiet room; monthly psychology review	Biosphere 2: crew split into two factions; prevention is key
Monotony	Rotating task assignments; hobby supplies; musical instruments; virtual reality recreation; celebration of holidays and milestones	HI-SEAS: 40 tasks/week maintained engagement; celebrations important
Circadian disruption	Strict light scheduling (blue-light curfew at 20:00); dedicated sleep cabins with blackout; morning bright-light exposure	Concordia + MARS-500: circadian disruption is universal in isolation
Loss of purpose	Clear mission objectives; science publication goals; shared responsibility for food production; mentoring roles	ISS: crew who feel scientifically productive report higher satisfaction

Stressor	Mitigation	Evidence
Health anxiety	Regular medical monitoring; telemedicine access; comprehensive pharmacy; cross-training in basic medical skills for all crew	All analogues: medical training for non-doctors is essential

9.3 Crew Selection Criteria

Based on analogue research, the 4-person crew should include:

Role	Skills	Backup Competence
Commander / Engineer	Leadership, mechanical/electrical engineering	Basic agriculture, first aid
Flight Surgeon / Biologist	Medicine, biology, psychology	Laboratory techniques, emergency repair
Agricultural Scientist	Botany, aquaculture, food science	Water systems, composting
Geologist / ISRU Specialist	Geology, chemistry, robotics	EVA specialist, communication systems

All crew must be cross-trained in: emergency medicine, fire suppression, EVA operations, basic repair of all critical systems.

10. Safety and Emergency Systems

10.1 Emergency Scenarios and Responses

Scenario	Detection	Response	Shelter Time
Rapid depressurisation	Pressure sensors (< 65 kPa alarm)	Seal affected zone; don emergency masks; locate and patch breach	Minutes
Fire	Smoke + thermal sensors	CO ₂ extinguishers; seal compartment; ventilate after extinguishing	Minutes-hours
Solar particle event	Space weather alert (8 min warning from Earth)	Crew to storm shelter; remain until all-clear	24-72 hours
Dust storm (major)	Atmospheric opacity sensors	Load priority mode; LED agriculture; battery conservation	Weeks

Scenario	Detection	Response	Shelter Time
Medical emergency	Crew report; vital monitors	Stabilise; telemedicine consultation; surgical intervention if trained	—
Communication loss	Signal timeout	Autonomous operations per standing protocols; antenna diagnostic	Days-weeks
Water system failure	Flow + quality sensors	Switch to emergency reservoir (10 m ³ = 100 days at reduced use)	Days
Crop failure	Visual + sensor monitoring	Emergency rations (6 months stored); accelerated replanting	Weeks-months

10.2 Emergency Consumables

Item	Quantity	Duration (4 crew)
Emergency food rations	2,400 kg (freeze-dried)	180 days
Water reserve (underfloor)	10,000 L	100 days (at 25 L/person/day)
Oxygen candles (backup)	200 units	60 days
Medical supplies	Full pharmacy	2 years
Spare parts (critical systems)	Comprehensive kit	—

11. Updated Specification Summary

Parameter	v1 (2024)	v2.0 (2026)	Notes
Interior height	3 m	5 m	Mezzanine, vertical farming, psychology
Floor area	1,000 m ²	1,000 m² + 400 m² mezzanine	Total usable: ~1,400 m ²
Interior volume	3,000 m ³	5,000 m³	+67%
Climate zones	1	3	Warm / Temperate / Cool
Agricultural area (effective)	200 m ²	~1,200 m²	400 m ² base × 3 tiers
Algae bioreactors	20 m ²	40 m²	Doubled capacity
Aquaponics	—	30 m² fish tanks + grow beds	Tilapia + vegetables
Airlocks	1	2 + emergency hatch	Redundancy

Parameter	v1 (2024)	v2.0 (2026)	Notes
Radiation shielding	UV coating	Regolith + water wall + storm shelter	Multi-layered
Water system	Not specified	Closed-loop, $\geq 93\%$ recovery + 10 m³ reserve	ISS-derived + plant transpiration
Waste recycling	Not specified	Full waste-to-resource cycle	Compost, struvite, biochar, biogas
Energy storage	Not specified	200 kWh LFP + RHU backup	72 h critical autonomy
ISRU capability	Not specified	Atmospheric + regolith processing	CO ₂ →CH ₄ , silica, iron, fertiliser
Crew privacy	Open plan	4 cabins (12 m²) + quiet room	Acoustic insulation
Medical bay	Not specified	15 m² equipped facility	STC ≥ 35 Ultrasound, pharmacy, telemedicine
Storm shelter	Not specified	20 m² with ≥ 2 m regolith	72 h capacity for 4 crew
Frame weight	5,500 kg	~8,500 kg	+55%
Glass area (walls)	~420 m ²	~700 m²	+67%
Crew routines	Not specified	Full daily/weekly/monthly/annual schedules	Evidence-based
Simulation framework	Not specified	5-stage validation cycle per domain	Investigation → Assessment

12. Updated Cost Estimation

12.1 Materials Cost (Excluding Transport)

Component	v1 Cost	v2.0 Estimated Cost	Notes
Silicone rubber + foam	\$53,500	~\$75,000	More foam for partitions + composting containers
Titanium/Inconel frame	\$165,000	~\$255,000	+55% weight for 5 m height
Borosilicate glass	\$500,000	~\$835,000	+67% wall area
Solar panels	\$200,000	~\$200,000	Roof area unchanged
Airlocks (2 + hatch)	\$100,000	~\$210,000	2 units + emergency hatch
Battery system (200 kWh LFP)	—	~\$160,000	Upgraded from v2 spec
Water recycling system	—	~\$200,000	ISS-grade VCD + biofilm reactor
LED grow lights (1,200 m ²)	—	~\$120,000	Full-spectrum agricultural LEDs

Component	v1 Cost	v2.0 Estimated Cost	Notes
Aquaponics system	—	~\$50,000	Tanks, biofilters, plumbing
Waste processing system	—	~\$80,000	Composting + vermicomposting + pyrolysis
ISRU equipment	—	~\$400,000	Sabatier + electrolysis + Haber-Bosch
Medical bay equipment	—	~\$150,000	Ultrasound, DXA, pharmacy, defibrillator
Storm shelter (reinforcement)	—	~\$50,000	HDPE panels + reinforced structure
Emergency supplies (6 months)	—	~100,000 <i>Freeze – dried food, O₂ candles, spares</i>	
Sensors + automation	—	~\$200,000	Environmental monitoring, digital twin
Total (materials)	\$1,018,500	~\$3,085,000	+203%

12.2 Transport Cost Impact

At SpaceX Starship target rate of ~\$200/kg to LEO (with Mars transit multiplier ~\$2,700/kg total):

Item	Mass (kg)	Transport Cost
Frame + glass + airlocks	~27,400	~\$74 M
All other systems	~8,600	~\$23 M
Emergency supplies	~3,000	~\$8 M
Total transport	~39,000	~\$105 M
Total project (materials + transport)		~\$108 M

Key insight: ISRU capability (\$400 K on Earth) can save tens of millions in future transport costs by producing replacement glass, rubber, and metals on Mars. The ISRU system pays for itself if it displaces even 2,000 kg of shipped materials.

13. Comparison with Earth Analogues

Feature	Biosphere 2	MARS-500	HI-SEAS	ISS	Mars RGH v2.0
Enclosed area	12,700 m ²	550 m ³	~70 m ²	~388 m ³	1,000 m² (5,000 m³)
Crew	8	6	6	3-6	4
Duration	2 years	520 days	Up to 1 year	6-12 months	Indefinite
Food self-sufficiency	83%	0% (stored)	0% (stored)	0% (stored)	Target: ≥ 80%

Feature	Biosphere 2	MARS-500	HI-SEAS	ISS	Mars RGH v2.0
Water recycling	Partial (wetlands)	Partial	None	93% (WRS)	Target: ≥ 95% Full cycle
Waste recycling	Composting + wetlands	None	None	Partial (water only)	
ISRU	N/A	N/A	N/A	N/A	
Radiation shielding	N/A	N/A	N/A	Partial (ISS hull)	Atmo-spheric + regolith Multi-layer
Gravity	1 g	1 g	1 g	~0 g	0.38 g

14. Risk Matrix

Risk	Probability	Impact	Mitigation	Residual Risk
Crop failure (disease/pest)	Medium	Critical	Species diversity; sealed growing chambers; biological pest control; 6-month food reserve	Low
Water system failure	Low	Critical	Dual-redundant pumps; 10 m ³ reservoir; manual purification capability	Low
Solar storm (SPE)	Low	High	Storm shelter; 8-minute warning from Earth; autonomous shelter protocol	Low
Dust storm (> 4 weeks)	Medium	High	200 kWh battery; LED agriculture; load prioritisation	Medium
Crew conflict	Medium	Medium	Private cabins; conflict training; monthly psych review; quiet room	Low

Risk	Probability	Impact	Mitigation	Residual Risk
Structural breach	Very Low	Critical	Pressure monitoring; self-sealing silicone patches; emergency masks; zone isolation	Low
Medical emergency	Low	High	Cross-trained crew; telemedicine; comprehensive pharmacy; surgical capability	Medium
ISRU failure	Medium	Low	ISRU is supplementary, not critical; all essential supplies stocked	Very Low

15. Conclusion

Version 2.0 of the Mars Rubber-Glass House transforms the original concept from a structural feasibility study into a comprehensive design for indefinite human habitation on Mars. The key advances are:

1. **From shelter to ecosystem** — The closed-loop life support system with waste-to-resource cycling creates a micro-biosphere where human waste feeds crops that feed humans, approaching the ideal of a self-sustaining system.
2. **From blueprint to operations manual** — Structured daily, weekly, monthly, and annual routines, informed by decades of analogue experience, address the human factors that make or break isolated communities.
3. **From Earth-dependent to Mars-native** — The ISRU strategy progressively reduces dependence on Earth resupply, consistent with the Three-Step Evolution roadmap toward autonomous structures (DOI: 10.5281/zenodo.16540444).
4. **From theoretical to validated** — The five-stage simulation methodology ensures every system is continuously investigated, modelled, simulated, validated, and scientifically assessed.

The approximately \$3 M materials cost (or ~\$108 M including transport) positions the Mars Rubber-Glass House as a serious candidate for early Martian settlement. As ISRU capabilities mature and transport costs decline — SpaceX targets \$200/kg to LEO within the decade — the economics become increasingly favourable.

“A house is not a home until it is lived in. On Mars, a house is not a home until it can sustain the lives within it — indefinitely, gracefully, and with dignity.”

16. Next Steps

1. **Earth Prototype** — Construct a full-scale v2.0 prototype on Earth (~\$3 M) to validate structural, agricultural, and life support systems
 2. **Digital Twin** — Develop a comprehensive simulation model integrating all subsystems for scenario testing
 3. **Closed-Loop Testing** — Conduct a 6-month sealed-habitat trial with 4 crew at the Earth prototype
 4. **ISRU Validation** — Test regolith processing with JSC Mars-1 simulant
 5. **Peer Review** — Submit design for review by materials scientists, life-support engineers, and psychologists
 6. **Zenodo v2 Release** — Publish all updated documents with cross-referenced DOIs
 7. **Integration with Mars Field Laboratories** — Plan robotic pre-deployment for regolith harvesting and site preparation (DOI: 10.5281/zenodo.15455156)
 8. **Phase 2 Transition** — Begin designing the biomimetic silica-aerogel upgrade path per the Three-Step Evolution roadmap
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17. References

1. NASA Baseline Values and Assumptions Document (BVAD), NASA/TP-2015-218570
 2. Hassler, D. M. et al. (2014). Mars' Surface Radiation Environment Measured with the Mars Science Laboratory. *Science*, 343(6169).
 3. Wheeler, R. M. (2017). Agriculture for Space: People and Places Paving the Way. *Open Agriculture*, 2(1), 14–32.
 4. Nelson, M. et al. (2009). The water cycle in closed ecological systems: Perspectives from the Biosphere 2 and Laboratory Biosphere systems. *Advances in Space Research*, 44(12), 1404–1412.
 5. Allen, J. P. (1991). *Biosphere 2: The Human Experiment*. Penguin Books.
 6. Basner, M. et al. (2013). Mars 520-d mission simulation reveals protracted crew hypokinesia. *PNAS*, 110(7), 2635–2640.
 7. Binsted, K. et al. (2013). HI-SEAS as an opportunity for long duration instrument/protocol testing. University of Hawai'i at Mānoa.
 8. Salam, A. (2009). The coldest job on earth. *BMJ*, b2453.
 9. Walford, R. L. et al. (1996). "Biospheric medicine" as viewed from the two-year first closure of Biosphere 2. *Aviation, Space, and Environmental Medicine*, 67(7), 609–617.
 10. Harrison, A. A. et al. (1991). *From Antarctica to Outer Space: Life in Isolation and Confinement*. Springer.
 11. Weyer, C. et al. (2000). Energy metabolism after 2 y of energy restriction: The Biosphere 2 experiment. *American Journal of Clinical Nutrition*, 72(4), 946–953.
 12. Haeuplik-Meusburger, S. et al. (2017). Habitability Studies and Full Scale Simulation Research: Preliminary Themes Following HISEAS Mission IV. *ICES Proceedings*.
 13. Massinger, R. A. (2024). Mars Rubber-Glass House Project. Zenodo. DOI: 10.5281/zenodo.16493055
 14. Massinger, R. A. (2024). Three-Step Evolution Towards Biomimetic and Self-Growing Structures. Zenodo. DOI: 10.5281/zenodo.16540444
 15. Massinger, R. A. (2025). Mars Field Laboratories. Zenodo. DOI: 10.5281/zenodo.15455156
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